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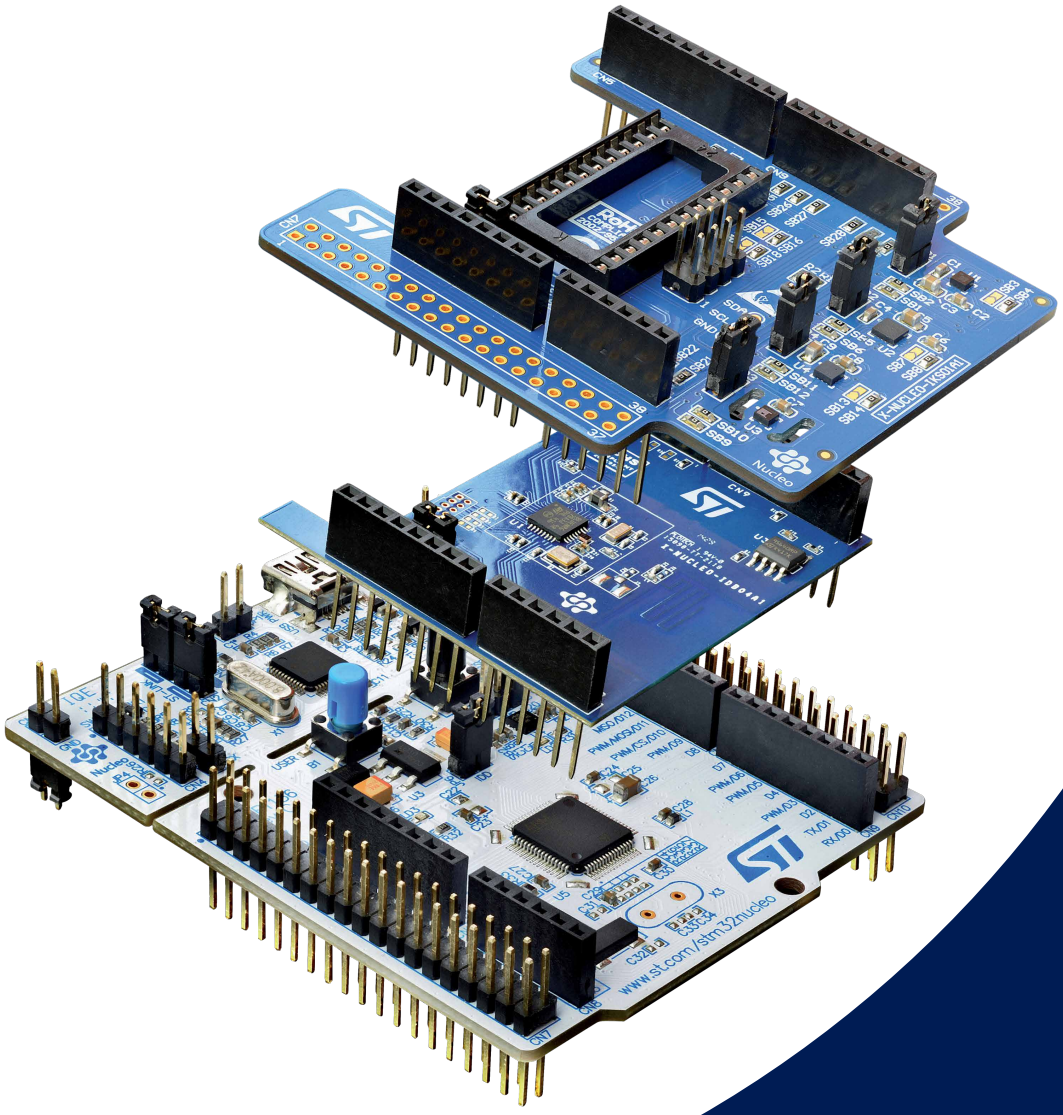
STM32
Open Development Environment



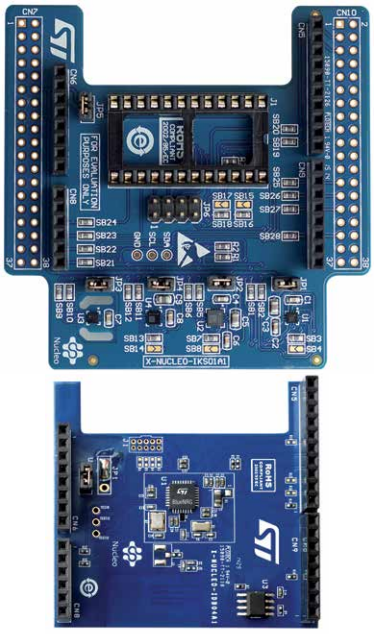
STM32 Open Development Environment

Fast, affordable
Development and prototyping

The STM32 Open Development Environment is a fast and affordable way to develop and prototype innovative devices and applications with state-of-the-art ST components leveraging the STM32 32-bit microcontroller family and a comprehensive set of functions for sensing, connectivity, power, audio, motor control and more. The combination of a broad range of expandable boards based on leading-edge commercial products and modular software, from driver to application level, enables fast prototyping of ideas that can be smoothly transformed into final designs.

STM32 Nucleo

To start your design, choose the appropriate STM32 Nucleo development board (MCU) and expansion (X-NUCLEO) boards (sensors, connectivity, audio, motor control etc.) for the functionality you need.
(Refer to www.st.com/stm32ode for details of board availability and out-of-the-box compatibility.)



STM32 Cube

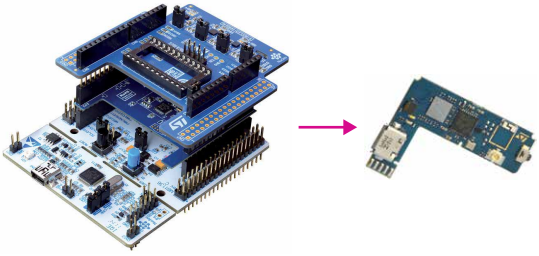
Next select your **development environment** (IAR EWARM, Keil MDK, and GCC-based IDEs) and use the free **STM32Cube tools and software.**

Download all the necessary software to run the functionality on the selected STM32 Nucleo expansion boards.

IAR SYSTEMS · KEIL Tools by ARM · GCC

Compile your design and upload it to the STM32 Nucleo development board. Then start developing and testing your applications.

Software developed on the STM32 Open Development Environment prototyping hardware can be directly used in an end product design using the same commercial ST components, or components from the same family as those found on the STM32 Nucleo boards.



All that you need

The STM32 Open Development Environment consists of a set of stackable boards and a modular open software environment designed around the STM32 microcontroller family.

Developer community and support
Online communities, development tools, documentation and user guides

STM32 Cube

STM32Cube software

A set of free tools and embedded software bricks to enable fast and easy development on the STM32, including a Hardware Abstraction Layer and middleware bricks.

STM32Cube expansion software

Expansion software provided free for use with the STM32 Nucleo expansion board and fully compatible with the STM32Cube software framework. It provides abstracted access to expansion board functionality through high-level APIs and sample applications.

STM32 Nucleo development boards

A comprehensive range of affordable development boards for all the STM32 microcontroller series, with unlimited unified expansion capabilities and integrated debugger/programmer functionality.

STM32 Nucleo expansion boards (X-NUCLEO)

Boards with additional functionality that can be plugged directly on top of the STM32 Nucleo development board or stacked on another expansion board.

STM32 Nucleo

MULTIPLE DEVELOPMENT ENVIRONMENTS

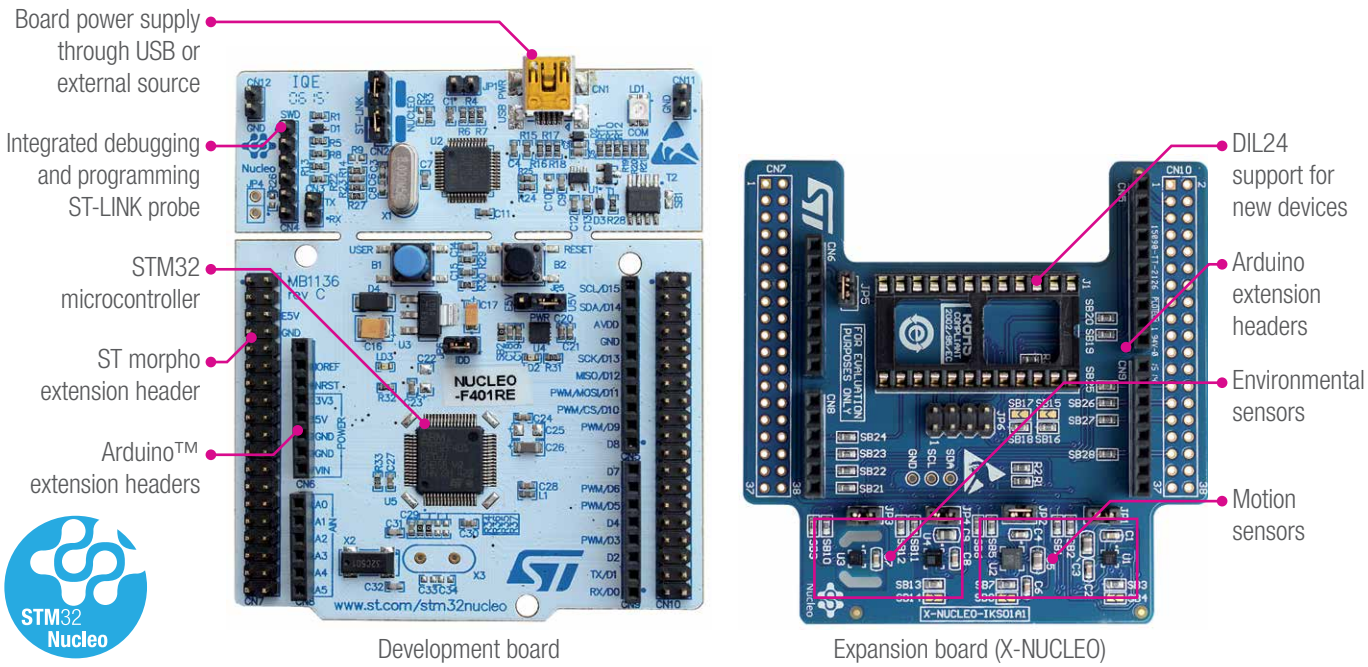
The STM32 Open Development Environment is compatible with a number of IDEs including IAR EWARM, Keil MDK, and GCC-based environments. Users can choose from three IDEs from leading vendors, which are free of charge and deployed in close cooperation with ST. These include Eclipse-based IDEs such as Ac6 System Workbench for STM32 and the MDK-ARM environment.

IAR SYSTEMS
KEIL Tools by ARM
GCC

STM32 Nucleo boards

THE SKY IS THE LIMIT

Sharing Arduino™ connectors and ST morpho headers, STM32 Nucleo boards can easily be extended with a large number of expansion boards available from ST and from third parties. Stack as many boards as you need to create the functionality required.



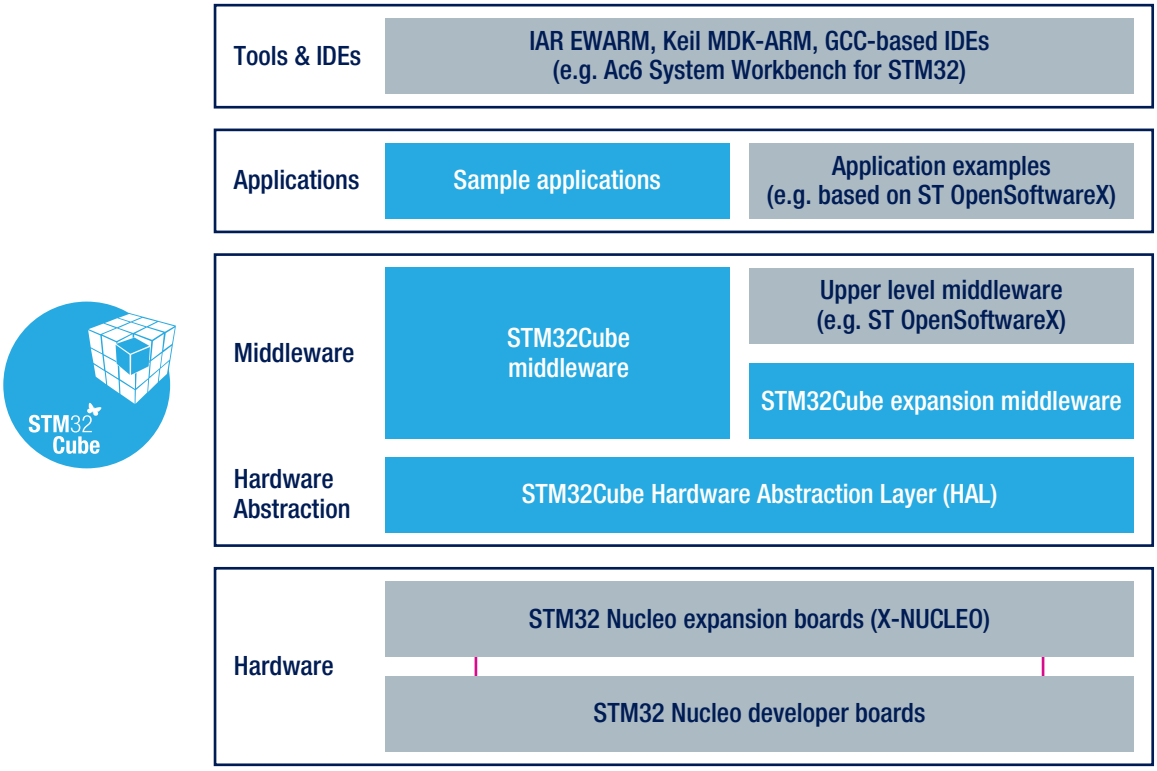
Development board

Expansion board (X-NUCLEO)

What you want to do	What we provide	Components	Availability
Process	Ultra-low power	STM32 L0 - ARM® Cortex®-M0+ ultra-low power 32-bit MCU	Available
	High performance	STM32 F401 - ARM® Cortex®-M4 high-performance 32-bit MCU	Available
Sense motion, pressure, humidity, temperature, distance, light, sound	Motion sensors & Environmental sensors	LSM6DS0 3-axis accelerometer + 3-axis gyroscope & LIS3MDL 3-axis magnetometer, HTS221 humidity and temperature & LPS25HB pressure	Available
		LSM6DS3 3-axis accelerometer + 3-axis gyroscope (add-on module on X-NUCLEO)	Available
	Proximity sensors	VL6180X FlightSense™ proximity, gesture and ambient light sensor	Available
	Microphone	MP34DT01-M digital microphone	Available
Connect wireless or wired	Bluetooth Low Energy 4.0	BlueNRG Bluetooth Low Energy (V4.0) network coprocessor	Available
	Bluetooth Low Energy 4.1	BlueNRG-MS based Bluetooth Low Energy (V4.1) Module	Available
	Wi-Fi	Wi-Fi module	H2 2015
	Sub-GHz radio	SPIRIT1 RF SPSGRF-915 and SPSGRF-868 modules	Available
	NFC	M24SR Dynamic NFC tag	Available
		M24LR Dynamic NFC tag	H2 2015
		NFC Reader	H2 2015
Move/Actuate	Motor driver	L6474 Stepper motor driver	Available
		powerSTEP01 High power stepper motor driver	Available
		Microstepping driver, DC motor driver, 2-axes motor driver	H2 2015
	Actuator	Protected switches for load ON/OFF control and load/device diagnostic	Q1 2016
Power/Drive	Energy management & battery	Battery and power management components	H2 2015
	LED Lighting	Single channel LED driver with integrated boost controller	H2 2015
	Audio processing	High-efficiency digital audio system	H2 2015
Translate signal conditioning	OpAmp	Operational Amplifiers	H2 2015
	Industrial Input/Output	CLT01 Protected digital termination array and VNI8200XP smart power solid state relay	H2 2015

STM32Cube
Development Software

STM32Cube is a set of free of charge tools and embedded software bricks to enable fast and easy development on the STM32 which simplifies and speeds up developers' work. **www.st.com/stm32cube**
The embedded software bricks include a Hardware Abstraction Layer (HAL) for easy porting from one STM32 device to another and middleware bricks for the most common functions (such as RTOS, USB , file system, TCP/IP stack, touch sensing or graphics).
A large number of code use examples are also included making it even easier to get started.



EXPANSION SOFTWARE

All STM32 Nucleo expansion boards come with STM32Cube expansion middleware. The middleware consists of source code drivers and sample applications built on top of the STM32Cube HAL, which provides abstracted access to board functionality through high-level APIs.

OPEN LICENSE MODELS

STM32Cube software and sample applications are covered by a mix of fully open source BSD license and ST licenses with very permissive terms.

Unleash your creativity
Develop your application

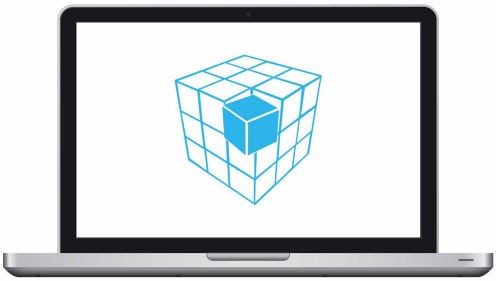
APPLICATION EXAMPLE: THREE STEP WIRELESS MULTI-SENSOR DEVICE

Choose the STM32 Nucleo boards

Select the STM32 Nucleo development board - in this case a NUCLEO-F401RE based on the STM32 F4.
www.st.com/stm32nucleo

Add the two expansion boards required: Environmental and motion sensors (X-NUCLEO- IKS01A1) and Bluetooth Low Energy (X-NUCLEO-IDB04A1).
www.st.com/x-nucleo

Use the STM32Cube software framework together with an IDE



IAR SYSTEMS

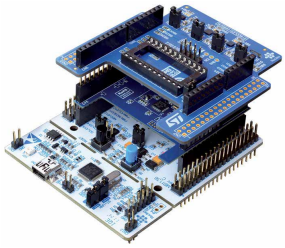
KEIL™ Tools by ARM

GCC

Download the BLUEMICROSYSTEM1 expansion software package (containing a pre-configured STM32Cube, expansion software and OpenSoftwareX SW) to get your application up and running quickly.
www.st.com/bluemicrosystem1

Obtain your free evaluation licence for the osxMotionFX sensor fusion software through the licensing wizard tool included in the package. Use your favorite development environment to compile the application and upload it to the boards.

Enjoy your newly created device



Download the **ST BlueMS** application from the Apple Store or Google Play store.
The application provides access to all the sensor data directly from your mobile device via the Bluetooth® Low Energy protocol.

OPEN SOFTWARE EXPANSION ENVIRONMENT (OpenSoftwareX)

Component drivers, middleware, and real use case SDKs are available on top of the STM32Cube software. Examples include sensor fusion and activity recognition libraries as well as a full-featured wireless sensor node SDK and a voice-over-Bluetooth Smart SDK. Refer to www.st.com/opensoftware for more information.

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